From a CIF File to a Crystal Shape

Every step from the numbers in a CIF file to the crystal shape, computed in your browser with the full working shown: lattice, metric tensor, centring and point group found from the atoms, structure factors, face spacings, the BFDH rule, the shape built plane by plane, and the map of every shape a cubic crystal can take.

Step 11 of 20 · next: **Morphology**

From a CIF File to a Crystal Shape, Step by Step

The panel asked whether a computer can start from the unit cell of an established MOF and produce the crystal shapes it can take. This page does the first, purely geometric version of that, in front of you: it reads a CIF file and computes every step from the six cell numbers to the finished shape, printing each intermediate number so the working can be checked by hand.

CIF file MOF-5 | HKUST-1 | UiO-66 | ZIF-8

CIF	LATTICE	SYMMETRY	WIDEST PLANES	SHAPE
276 atoms, 16.99 Å, 90°	**cubic, a = 16.991 Å**	**I-43m**	**{110}, d_{eff} = 12.01 Å**	**rhombic dodecahedron**

Computed in 67 ms in this browser by `mof_cif_shape.js`.

How to read this page

Each step has three parts. **In plain words** says what is being done and why. **The rule** gives the mathematics in general form. **With this file's numbers** applies the rule to the chosen CIF; everything in those boxes is computed when the page loads by `mof_cif_shape.js`, and nothing is typed in. For the four bundled files, a line marked **Python check** repeats the result of the independent Python implementation `code/morphology/run_cif_to_shape.py`, which also runs spglib. The test `website/tests/test_cif_to_shape.js` fails if the two disagree. The page prints in full from the browser, for reading on paper.

1 Read the File

IN PLAIN WORDS

A CIF is a plain text file. Two things in it matter here: the box that repeats to fill space (the unit cell, given by three edge lengths and three angles) and the list of atoms inside it. Each atom is given as an element and three **fractional coordinates**: how far along each edge of the box it sits, as a fraction of that edge.

THE RULE

Unit cell: a, b, c (Å) and α (between **b** and **c**), β (between **a** and **c**), γ (between **a** and **b**). An atom at fractional coordinates **x** = (x, y, z) sits at the point $\mathbf{r} = x\mathbf{a} + y\mathbf{b} + z\mathbf{c}$, and also at $\mathbf{r} + u\mathbf{a} + v\mathbf{b} + w\mathbf{c}$ for every whole number u, v, w.

WITH THIS FILE'S NUMBERS

```
_cell_length_a 16.991
_cell_length_b 16.991
_cell_length_c 16.991
_cell_angle_alpha 90
_cell_angle_beta 90
_cell_angle_gamma 90

loop_ (atoms)
C1 C 0.37705 0.0079 0.62295 0
C2 C 0.36851 0.89914 0.6875 0
H2 H 0.3777 0.8589 0.7234 0
C3 C 0.4061 0.0855 0.5939 0
… 272 more atom lines
```

The file lists **276 atoms** (C_{96} H_{120} N_{48} Zn_{12}), each by element and fractional coordinates (x, y, z), and the six cell numbers above. Its stated space group is `P 1`, that is, no symmetry is written down. Every symmetry used below is found from the numbers in the file.

2 The Lattice Vectors and the Metric Tensor

IN PLAIN WORDS

The six numbers are turned into three arrows, the cell edges **a**, **b**, **c**, written in ordinary x-y-z coordinates. From the arrows we build one 3×3 table, the metric tensor G, which holds every length and angle of the cell. Every later step that needs a distance or an angle reads it from G. G does not change when the crystal is turned, so it describes the lattice itself, not the way it happens to be drawn.

THE RULE

$\mathbf{a} = (a, 0, 0)$, $\mathbf{b} = (b\cos\gamma, b\sin\gamma, 0)$, $\mathbf{c} = (c\cos\beta, c(\cos\alpha - \cos\beta\cos\gamma)/\sin\gamma, c_z)$, with c_z fixed by $|\mathbf{c}| = c$.

Put the three edges as the rows of a matrix A. Then $G = A\,A^{\mathrm{T}}$, that is $G_{ij} = \mathbf{a}_i \cdot \mathbf{a}_j$: the diagonal holds a^2, b^2, c^2 and the off-diagonal entries $ab\cos\gamma$, $ac\cos\beta$, $bc\cos\alpha$.

Cell volume $V = \sqrt{(\det G)}$. The length of any lattice vector with integer or fractional components $\mathbf{u}$ is $|\mathbf{u}|^2 = \mathbf{u}^{\mathrm{T}}\, G\, \mathbf{u}$.

WITH THIS FILE'S NUMBERS

$\mathbf{a} = (a, 0, 0) = (16.9910, 0, 0)$

$\mathbf{b} = (b\cos\gamma, b\sin\gamma, 0) = (16.9910 \times 0.00000, 16.9910 \times 1.00000, 0) = (0.0000, 16.9910, 0.0000)$

$c_x = c\cos\beta = 16.9910 \times 0.00000 = 0.0000$

$c_y = c\,(\cos\alpha - \cos\beta\cos\gamma) / \sin\gamma = 16.9910 \times (0.00000 - 0.00000 \times 0.00000) / 1.00000 = 0.0000$

$c_z = \sqrt{(c^2 - c_x^2 - c_y^2)} = \sqrt{(288.694 - 0.000 - 0.000)} = 16.9910$

$$A = \begin{pmatrix} \mathbf{16.9910} & 0.0000 & 0.0000 \\ \mathbf{0.0000} & 16.9910 & 0.0000 \\ \mathbf{0.0000} & 0.0000 & 16.9910 \end{pmatrix} \text{ Å} \qquad G = A\,A^{\mathrm{T}} = \begin{pmatrix} \mathbf{288.694} & 0.000 & 0.000 \\ \mathbf{0.000} & 288.694 & 0.000 \\ \mathbf{0.000} & 0.000 & 288.694 \end{pmatrix} \text{ Å}^2$$

For example $G_{11} = \mathbf{a}\cdot\mathbf{a} = 16.9910^2 = 288.694$ and $G_{12} = \mathbf{a}\cdot\mathbf{b} = ab\cos\gamma = 16.9910 \times 16.9910 \times 0.00000 = 0.000$.

$\det G = G_{11}(G_{22}G_{33} - G_{23}^2) - G_{12}(G_{12}G_{33} - G_{23}G_{13}) + G_{13}(G_{12}G_{23} - G_{22}G_{13}) = 24060998$ Å^6

Volume $V = \sqrt{(\det G)} =$ **4905.20 Å³**

3 Is the Lattice Cubic? The Conventional Cell

IN PLAIN WORDS

Files often store the smallest box that repeats, not the most symmetric one. A face-centred cube, for example, can be stored as a slanted box with 60° angles that is one quarter of the cube. To see the cube we look for whole-number combinations of the stored edges that are equal in length and at right angles.

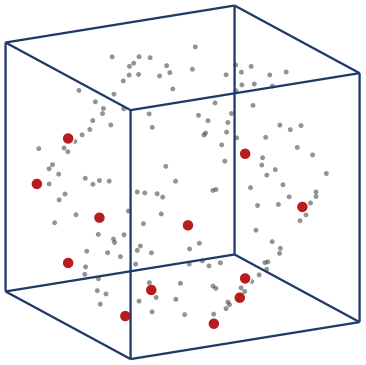
THE RULE

The lattice is cubic exactly when there is an integer matrix M (its rows say how to combine the stored edges) such that $M\,G\,M^{\mathrm{T}} = a_c^2\,I$. The new edges $M\,A$ form the conventional cube of edge a_c, which contains $|\det M|$ lattice points. Three candidates cover every cubic lattice: $M = I$ (the file already holds the cube); the 60° rhombohedral cell of a face-centred cube ($\det M = 4$); the 109.47° rhombohedral cell of a body-centred cube ($\det M = 2$). Agreement is tested to a relative tolerance of 2×10^{-4}, which allows for the rounding of the numbers in the file.

WITH THIS FILE'S NUMBERS

Here G is already a^2 times the identity: the three edges are equal (16.9910 Å) and at right angles (largest off-diagonal entry 0.0000 Å²). The cell in the file **is** the cube, with a_c = 16.9910 Å, and M is the identity. Whether further lattice points sit inside it (a centred cube) is decided from the atoms in Step 4.

Python check (`code/morphology/run_cif_to_shape.py`): conventional cell edge 16.9910 Å (transformation P); spglib standard cell 16.9910 Å.



Blue: the conventional cube, edge 16.991 Å. The file stores this cube directly. Red: metal atoms; grey: the other non-hydrogen atoms, all 276 atoms placed from the file's fractional coordinates. Drag to rotate.

4 Lattice Centring, Found from the Atoms

IN PLAIN WORDS

A cube of lattice points may have extra points that repeat the whole pattern: one at the centre of the body (a body-centred cube, I) or one at the centre of each face (a face-centred cube, F). We test this directly. Shift every atom by half a cell along a diagonal; if each one lands on an atom of the same element, that shift is a symmetry of the crystal.

THE RULE

A translation $\mathbf{t}$ is a symmetry when, for every atom j, some atom k of the same element satisfies $|(\mathbf{x}_j + \mathbf{t} - \mathbf{x}_k - \mathbf{m})\,A_c| < 0.02$ Å for some whole-number vector $\mathbf{m}$ (the wrap-around into the next cell). I-centring is $\mathbf{t}$ = (½, ½, ½); F-centring is the three vectors (0, ½, ½), (½, 0,

½), (½, ½, 0). The test is run with all atoms and again without hydrogen, because hydrogen positions in a CIF are often placed by software and can break a symmetry the framework has.

WITH THIS FILE'S NUMBERS

Atoms used	I: + (½, ½, ½)	F: + (0, ½, ½)	F: + (½, 0, ½)	F: + (½, ½, 0)
all 276	252 / 276	4 / 276	4 / 276	4 / 276
without H	156 / 156	4 / 156	4 / 156	4 / 156

Each entry counts the atoms j that have an atom of the same element within 0.02 Å of $\mathbf{x}_j + \mathbf{t}$. A translation is a symmetry only if every atom passes. Result: the lattice is **body-centred (I)**, found on the framework without hydrogen.

Hydrogen breaks the symmetry in this file

With every atom, (½, ½, ½) maps only 252 / 276 atoms; the ones that fail are hydrogen atoms whose positions in this file do not follow the framework. Without hydrogen all 156 atoms pass. The framework symmetry is used from here on, as spglib does for this file.

Python check (`code/morphology/run_cif_to_shape.py`): centring I, found without hydrogen; spglib uses hydrogen: no.

5 The Point Group, Found from the Atoms

IN PLAIN WORDS

Now we turn and reflect the crystal instead of shifting it. If the crystal looks exactly the same after a rotation (perhaps combined with a shift), that rotation is a symmetry. Which rotations survive decides which crystal faces are equivalent, and so which faces must grow at the same rate.

THE RULE

An operation (R, $\mathbf{t}$) is a symmetry when every atom j is sent onto an atom of the same element: $R\mathbf{x}_j + \mathbf{t} \equiv \mathbf{x}_k$ (mod 1, within 0.02 Å). The shift $\mathbf{t}$ is not known in advance. However, a symmetry must send the first atom of the rarest element onto some atom of that element, so only that many values of $\mathbf{t}$ need to be tried. Four operations decide the cubic point group: inversion, a 3-fold axis along the body diagonal, a 4-fold axis along an edge, and a 4-fold rotoinversion along an edge.

WITH THIS FILE'S NUMBERS

Operation	Acts as	R	Atoms mapped (best t)	t found	Symmetry?
inversion	(x, y, z) → (−x, −y, −z)	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	12 / 156	—	no
3-fold axis along [111]	(x, y, z) → (z, x, y)	$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$	156 / 156	(0, 0, 0)	yes
4-fold axis along [001]	(x, y, z) → (−y, x, z)	$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	12 / 156	—	no
4-fold rotoinversion along [001]	(x, y, z) → (y, −x, −z)	$\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	156 / 156	(½, ½, ½)	yes

Decision: a 3-fold axis is required for a cubic crystal. With inversion and a 4-fold axis the point group is **m-3m** (48 operations); with the 3-fold axis and the −4 axis but no inversion it is **-43m** (24 operations). Here: **-43m**.

Without inversion the eight {111} planes split into two sets of four that are not related by symmetry, $\{111\}_+$ ($hkl > 0$) and $\{111\}_-$ ($hkl < 0$). The same holds for every family with no zero index. If the two halves grew at different rates the crystal could become tetrahedral; BFDH gives them equal distances (Step 8), and the tetrahedral shapes are explored on the Morphology page.

Python check (`code/morphology/run_cif_to_shape.py`): point group -43m by the same search; spglib: I-43m (No. 217).

6 Which Planes Cancel: Structure Factors and the Space Group

IN PLAIN WORDS

A crystal face is a plane of atoms, and the face grows by adding the next identical layer. If the centring places an identical layer exactly half-way between two planes of a family, the true repeat distance of that family is half its nominal spacing. X-ray diffraction shows this directly: the waves scattered by the two layers are exactly out of step and cancel, so that reflection is missing. We compute this cancellation from the atoms in the file.

THE RULE

Structure factor: $F(hkl) = \Sigma_j Z_j \exp(2\pi i\,(hx_j + ky_j + lz_j))$, taking each atom's atomic number Z_j as its weight.

If the translations $\mathbf{t}_1, \ldots, \mathbf{t}_m$ are symmetries, the atoms come in sets $\{\mathbf{x}, \mathbf{x} + \mathbf{t}_1, \ldots\}$, and the sum factorises as $F(\mathbf{h}) = S(\mathbf{h}) \times \Sigma_{\text{one atom per set}} Z_j \exp(2\pi i\, \mathbf{h}\cdot\mathbf{x}_j)$, with $S(\mathbf{h}) = \Sigma_s \exp(2\pi i\, \mathbf{h}\cdot\mathbf{t}_s)$. When $S(\mathbf{h}) = 0$, the reflection vanishes whatever the atoms are. This is a **systematic absence**. The weights change the size of an allowed reflection but never make an absent one non-zero.

Glide planes (a reflection combined with a half-cell shift) forbid further reflections. Each possible glide group is checked against the one reflection it would remove.

WITH THIS FILE'S NUMBERS

For this lattice, $S(\mathbf{h}) = 1 + e^{i\pi(h+k+l)}$. Each term is +1 or −1, so S is 2 when $h + k + l$ is even, and 0 otherwise.

(*hkl*)	*S*(h), term by term	By centring	|*F*(*hkl*)| / Σ*Z*, from the 156 atoms
(100)	1 − 1 = 0	absent	**0.00000**
(200)	1 + 1 = 2	allowed	**0.20351**
(110)	1 + 1 = 2	allowed	**0.27944**
(220)	1 + 1 = 2	allowed	**0.13093**
(111)	1 − 1 = 0	absent	**0.00000**
(222)	1 + 1 = 2	allowed	**0.36197**
(210)	1 − 1 = 0	absent	**0.00000**
(420)	1 + 1 = 2	allowed	**0.04393**
(211)	1 + 1 = 2	allowed	**0.20265**
(422)	1 + 1 = 2	allowed	**0.12748**
(311)	1 − 1 = 0	absent	**0.00000**
(331)	1 − 1 = 0	absent	**0.00000**

Every reflection the centring forbids comes out as exactly zero from the atoms; the allowed ones do not (ΣZ = 1272, with Z the atomic number of each atom as its weight).

Centring I with point group -43m leaves 2 possible space groups. The others contain glide planes, and each glide forbids one more reflection:

- **I-43d** (No. 220): hhl: 2h + l = 4n, so (110) would vanish. From the atoms |F(110)|/ΣZ = 0.2794, not zero, so I-43d is ruled out.

Space group: I-43m (No. 217). For these groups the centring is the only rule that removes faces, so the first allowed order of (*hkl*) follows from the centring alone.

Python check (`code/morphology/run_cif_to_shape.py`): |F|/ΣZ for (100), (110), (111), (200): 0.00000, 0.27944, 0.00000, 0.20351; spglib: **I-43m**.

7 Face Families, Spacings and Effective Spacings

IN PLAIN WORDS

Every set of parallel lattice planes is a possible crystal face, labelled by three whole numbers (*hkl*), its Miller indices. Planes related by the crystal's symmetry form a family, written {*hkl*}, and all members of a family behave identically. For each family we need one number, the distance between neighbouring identical layers. The picture shows why the centring can halve it.

THE RULE

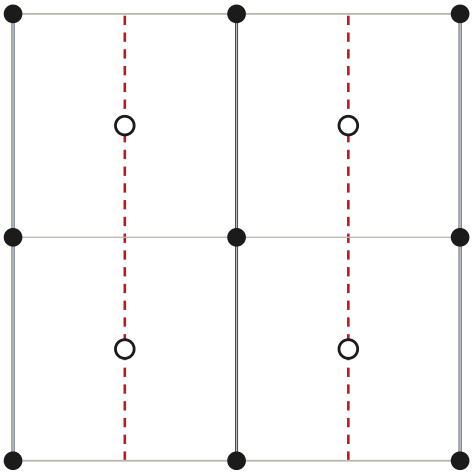
Spacing: $1/d^2 = \mathbf{h}^{\mathrm{T}}\, G^{-1}\, \mathbf{h}$. For the cubic cell $G = a^2 I$, so $d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$.

First allowed order: the smallest n for which n(*hkl*) is not absent (Step 6). Effective spacing $d_{\mathrm{eff}} = d_{hkl} / n$.

The members of a family are all signed permutations of (*hkl*) in point group m-3m. In -43m a family with no zero index splits into two halves of equal size, told apart by the sign of *hkl*. Every primitive (*hkl*) with indices up to 6 is listed.

Planes (100) (110)

(100) planes, viewed down [001]



Filled dots: lattice points at height 0; open dots: at height ½*a*. Two cells of edge *a* = 16.991 Å are shown. Blue lines: the planes (100) with spacing *d* = 16.991 Å. Red dashed lines: an identical layer of lattice points lies exactly half-way, so the true repeat is *d*/2 = **8.495 Å**. That is why (100) is absent and its first allowed order is (200).

WITH THIS FILE'S NUMBERS

#	Family	Planes	$h^2+k^2+l^2$	$d = a/\sqrt{(...)}$ (Å)	First allowed order	n	$d_{eff} = d/n$ (Å)
1	{110}	12	2	16.991 / √2 = 12.014	(110) allowed	1	12.014
2	{100}	6	1	16.991 / √1 = 16.991	(100) absent, (200) allowed	2	8.495
3	{211}₋	12	6	16.991 / √6 = 6.937	(211) allowed	1	6.937
4	{211}₊	12	6	16.991 / √6 = 6.937	(211) allowed	1	6.937
5	{310}	24	10	16.991 / √10 = 5.373	(310) allowed	1	5.373
6	{111}₋	4	3	16.991 / √3 = 9.810	(111) absent, (222) allowed	2	4.905
7	{111}₊	4	3	16.991 / √3 = 9.810	(111) absent, (222) allowed	2	4.905
8	{321}₋	24	14	16.991 / √14 = 4.541	(321) allowed	1	4.541
9	{321}₊	24	14	16.991 / √14 = 4.541	(321) allowed	1	4.541
10	{411}₋	12	18	16.991 / √18 = 4.005	(411) allowed	1	4.005
11	{411}₊	12	18	16.991 / √18 = 4.005	(411) allowed	1	4.005
12	{210}	24	5	16.991 / √5 = 7.599	(210) absent, (420) allowed	2	3.799

The 12 widest-spaced of 97 families with indices up to 6 are listed; show all 97 Every one of them is used in Step 9.

Python check (`code/morphology/run_cif_to_shape.py`): the widest families and d_{eff}: {110} 12.014, {100} 8.495, {211}$_-$ 6.936, {211}$_+$ 6.936, {310} 5.373, {111}$_-$ 4.905 Å.

8 How Far Out Each Face Sits: the BFDH Rule

IN PLAIN WORDS

A face made of widely spaced planes is densely packed with atoms and is flat at the molecular scale, so new material sticks to it slowly. A slow face moves outward slowly and stays large on the finished crystal, while fast faces grow themselves out of existence. The simplest version of this idea says that each face's distance from the centre is inversely proportional to its effective spacing.

THE RULE

BFDH rule (Bravais 1866, Friedel 1907, Donnay and Harker 1937): the distance of face family f from the centre is $R_f \propto 1/d_{\mathrm{eff},f}$. Scale it so the widest-spaced family is at distance 1: $R_f = d_{\mathrm{eff,max}} / d_{\mathrm{eff},f}$. Only ratios of distances matter, because the shape does not depend on its size.

WITH THIS FILE'S NUMBERS

The widest spacing is $d_{\mathrm{eff,max}}$ = 12.014 Å, for {110}. Dividing by it puts that family at distance 1:

Family	$d_{\mathrm{eff,max}} / d_{\mathrm{eff}}$	R_f	Exact value
{110}	12.014 / 12.014	1.0000	1
{100}	12.014 / 8.495	1.4142	√2
{211}$_-$	12.014 / 6.937	1.7321	√3
{211}$_+$	12.014 / 6.937	1.7321	√3
{310}	12.014 / 5.373	2.2361	√5
{111}$_-$	12.014 / 4.905	2.4495	√6
{111}$_+$	12.014 / 4.905	2.4495	√6
{321}$_-$	12.014 / 4.541	2.6458	√7

The exact values follow because every d_{eff} is $a/(n\sqrt{(h^2+k^2+l^2)})$: the cell edge a cancels, so R_f^2 is a ratio of whole numbers, $R_f^2 = n_f^2(h^2+k^2+l^2)_f \,/\, n_0^2(h^2+k^2+l^2)_0$. So the BFDH shape depends only on the centring and the point group, not on the size of the cell or on the atoms.

Python check (`code/morphology/run_cif_to_shape.py`): BFDH distances {110} 1.0000, {100} 1.4142, {211}$_-$ 1.7321, {211}$_+$ 1.7321.

9 Building the Shape, Plane by Plane

IN PLAIN WORDS

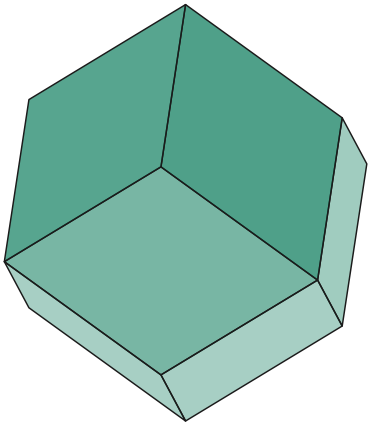
Every member of every family becomes a flat wall at its distance R from the centre, facing outward. The crystal is everything that lies inside all the walls. We build it as a sculptor would: start from a large block and cut it with one wall at a time, nearest walls first. Most walls end up outside the shape and cut nothing. The animation shows only the cuts that change it.

THE RULE

The shape is the convex polytope $P = \{\mathbf{x} : \mathbf{n}_p \cdot \mathbf{x} \le R_p$ for every plane $p\}$, with outward unit normal $\mathbf{n} = \mathbf{h}/|\mathbf{h}|$. In a cubic cell the normal of (hkl) points along (h, k, l), and the wall is the plane $hx + ky + lz = R|\mathbf{h}|$.

One cut: every face polygon is clipped to the inner side of the new plane (each edge that crosses the plane is cut at the crossing point $\mathbf{A} + (\mathbf{B} - \mathbf{A})\, s_A/(s_A - s_B)$, with $s = \mathbf{n}\cdot\mathbf{x} - R$). The crossing points, ordered around their centre, form the new face.

A corner of the shape is a point on three planes with independent normals: the 3×3 linear system $N\mathbf{x} = \mathbf{r}$, solved by Cramer's rule.



Start. The 12 planes of the nearest family, {110} at R = 1.0000, already enclose a closed shape: **rhombic dodecahedron**, 12 faces. The remaining **1718** planes all lie outside this shape (Step 10 shows by how much), so this is the final crystal.

Face colours: blue {100}, red {111} (dark red: the second half {111}_ where the point group splits it), green {110}. Drag to rotate.

WITH THIS FILE'S NUMBERS

One corner of the final shape, worked by hand. It lies on the three planes

(110): $x + y = R|\mathbf{h}| = 1.0000 \times 1.4142 = 1.4142$

($1\bar{1}0$): $x - y = R|\mathbf{h}| = 1.0000 \times 1.4142 = 1.4142$

(101): $x + z = R|\mathbf{h}| = 1.0000 \times 1.4142 = 1.4142$

$$N = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} 1.4142 \\ 1.4142 \\ 1.4142 \end{pmatrix}, \quad \det N = -2$$

$x = \det N_x / \det N = -2.8284 / -2 =$ **1.4142** (N_x is N with column 1 replaced by $\mathbf{r}$)

$y = \det N_y / \det N = 0.0000 / -2 =$ **0.0000** (N_y is N with column 2 replaced by $\mathbf{r}$)

$z = \det N_z / \det N = 0.0000 / -2 =$ **0.0000** (N_z is N with column 3 replaced by $\mathbf{r}$)

So the corner is at (1.4142, 0.0000, 0.0000). It must also satisfy every other plane: of all 1730 planes, the largest value of $\mathbf{n}\cdot\mathbf{x} - R$ at this point is 0.000000 (zero or below means inside), and 4 planes pass exactly through it.

10 Which Faces Survive, and by How Much the Others Miss

IN PLAIN WORDS

For each family we compare two distances: how far out its wall sits, and how far the finished shape reaches in that direction. If the wall is farther out than the shape reaches, it never touches the crystal and that face does not appear.

THE RULE

Reach of the shape along a direction **n** (the support function): $h_P(\mathbf{n})$ = max over the corners **v** of **n**·**v**. On the finished shape every wall satisfies $R_f \geq h_P(\mathbf{n}_f)$. A family carries faces when equality holds and the wall meets the shape in a polygon of positive area; it is absent when $R_f > h_P(\mathbf{n}_f)$, and the difference is the margin by which it misses. When $R_f = h_P(\mathbf{n}_f)$ but the face has zero area, the wall touches the shape only at a corner or along an edge.

WITH THIS FILE'S NUMBERS

Family	Its distance R_f	Reach of the shape $h_P(\mathbf{n}_f)$	Result
{110}	**1.0000**	**1.0000**	**face**, 12 polygons, 100.0% of the area
{100}	**1.4142**	**1.4142**	**touches** the shape at corners only, zero area
{211}$_-$	**1.7321**	**1.1547**	misses by 0.5774
{211}$_+$	**1.7321**	**1.1547**	misses by 0.5774
{310}	**2.2361**	**1.3416**	misses by 0.8944
{111}$_-$	**2.4495**	**1.2247**	misses by 1.2247
{111}$_+$	**2.4495**	**1.2247**	misses by 1.2247
{321}$_-$	**2.6458**	**1.1339**	misses by 1.5119
{321}$_+$	**2.6458**	**1.1339**	misses by 1.5119
{411}$_-$	**3.0000**	**1.3333**	misses by 1.6667
{411}$_+$	**3.0000**	**1.3333**	misses by 1.6667
{210}	**3.1623**	**1.2649**	misses by 1.8974
{332}$_-$	**3.3166**	**1.2060**	misses by 2.1106
{332}$_+$	**3.3166**	**1.2060**	misses by 2.1106

Of the 97 families, 1 carry faces. For the other 96, the smallest gap between the plane and the shape is 0.5774, apart from {100}, which touches exactly.

A shape exactly on a boundary

The {100} planes pass exactly through corners of the shape (R = 1.4142 = h_P). Any growth rate for these faces slightly slower than BFDH would make them appear as small faces at those corners. For a cubic I-centred lattice this happens for every cell size, because $R_{100}/R_{110} = \sqrt{2}$ exactly.

Python check (`code/morphology/run_cif_to_shape.py`): 97 families with SciPy half-space intersection on the ten widest; the narrower 87 miss that shape by at least 1.6667.

11 Measuring and Naming the Shape

IN PLAIN WORDS

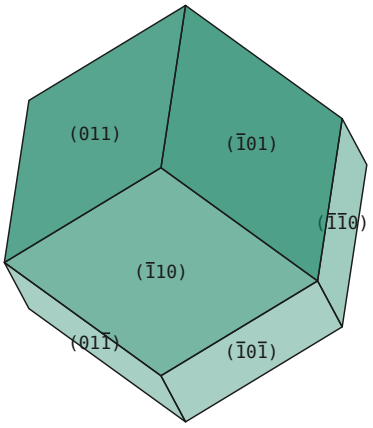
The finished shape is reduced to a few numbers: how much of its surface each family covers, its volume, and how close it is to a sphere. The name follows from which families are present and from the outline of their faces.

THE RULE

Area of a face with corners $\mathbf{v}_1, \ldots, \mathbf{v}_m$ in order: $A = \tfrac{1}{2} \left|\Sigma_{t=2}^{m-1} (\mathbf{v}_t - \mathbf{v}_1) \times (\mathbf{v}_{t+1} - \mathbf{v}_1)\right|$.

Volume: the shape is a union of pyramids from the centre to each face, of height R_f, so $V = \tfrac{1}{3} \Sigma_f R_f A_f$.

Sphericity: $\Psi = \pi^{1/3} (6V)^{2/3} / A$, the area of the sphere of the same volume divided by the actual area (1 for a sphere).



The finished BFDH shape for this file, with Miller indices on the front faces. Drag to rotate.

WITH THIS FILE'S NUMBERS

A {110} face, (110), is a polygon with 4 corners:

$\mathbf{v}_1$ = (1.4142, 0.0000, 0.0000)

$\mathbf{v}_2$ = (0.7071, 0.7071, −0.7071)

$\mathbf{v}_3$ = (0.0000, 1.4142, 0.0000)

$\mathbf{v}_4$ = (0.7071, 0.7071, 0.7071)

Area = $\tfrac{1}{2}|\Sigma_t (\mathbf{v}_t - \mathbf{v}_1) \times (\mathbf{v}_{t+1} - \mathbf{v}_1)| = \tfrac{1}{2}|(2.0000, 2.0000, 0.0000)|$ = **1.4142**, and 12 such faces give 16.9706.

Total area A = 16.9706 = **16.9706**; area fractions {110} 1.0000.

Volume $V = \tfrac{1}{3} \Sigma_f R_f A_f$ (a pyramid from the centre to each face) = ⅓ × 1.0000 × 16.9706 = **5.6569**

Sphericity $\Psi = \pi^{1/3}(6V)^{2/3}/A$ = 1.4646 × $(6 \times 5.6569)^{2/3}$ / 16.9706 = **0.9047** (a sphere has 1, a cube 0.806).

Counts: 12 faces, 14 corners, 24 edges (Euler: $F + V - E = 2$).

Name: **rhombic dodecahedron**. Only the twelve {110} faces are present, each a rhombus. A closed-form check by hand: the diagonals of each rhombus are 2.0000 and 1.4142 (ratio 1.4142 = √2), so each face has area ½ × 2.0000 × 1.4142 = 1.4142 and twelve give 16.9706; the volume of a rhombic dodecahedron with face distance 1 is 4√2 = 5.6569.

Python check (`code/morphology/run_cif_to_shape.py`): rhombic dodecahedron, 12 faces, 14 corners, Ψ = 0.9047, area fractions {110} 1.0000.

12 Beyond BFDH: Every Shape a Cubic Crystal Can Take

IN PLAIN WORDS

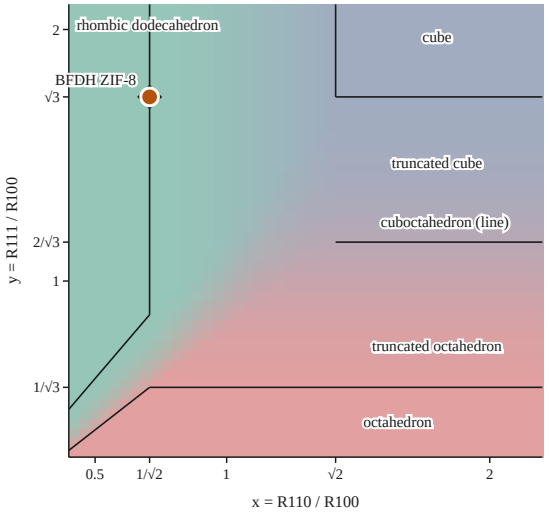
BFDH fixes one set of distances from the geometry alone. Real growth conditions (solvent, temperature, added modulators) change how fast each face grows, which moves the distances. For a cubic crystal bounded by {100}, {110} and {111}, two ratios decide the shape completely, so all possible shapes fit on one map. Click or drag on the map to see the shape at any point. The diamond marks where BFDH puts this file.

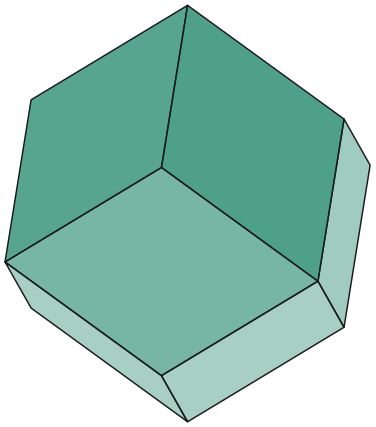
THE RULE

Set $R_{100} = 1$, $x = R_{110}/R_{100}$, $y = R_{111}/R_{100}$. Each boundary follows from the reach of one simple shape (Step 10):

- **Cube** (six {100} walls at 1, corners (±1, ±1, ±1)): its reach along (111)/√3 is 3/√3 = √3, and along (110)/√2 it is 2/√2 = √2. No other face cuts it when $y \geq \sqrt{3}$ and $x \geq \sqrt{2}$.
- **Octahedron** (eight {111} walls at y, corners $(\pm\sqrt{3}\,y, 0, 0)$ and permutations): its reach along (100) is $\sqrt{3}\,y$, and along (110)/√2 it reaches the edge midpoint $(\sqrt{3}y/2, \sqrt{3}y/2, 0)$, which gives $\sqrt{(3/2)}\,y$. It is pure when $y \leq 1/\sqrt{3}$ and $x \geq \sqrt{(3/2)}\,y$.
- **Rhombic dodecahedron** (twelve {110} walls at x, corners $(\pm\sqrt{2}\,x, 0, 0)$ and $(\pm x/\sqrt{2})(1, 1, 1)$): its reach along (100) is $\sqrt{2}\,x$ and along (111)/√3 it is $\sqrt{(3/2)}\,x$. It is pure when $x \leq 1/\sqrt{2}$ and $y \geq \sqrt{(3/2)}\,x$.
- **Cuboctahedron**: the {111} triangles and {100} squares share corners at (1, 1, 0), where (1,1,1)·(1,1,0)/√3 = 2/√3. So it occurs only on the line $y = 2/\sqrt{3}$, and only for $x \geq \sqrt{2}$. Above the line the shape is a truncated cube, below it a truncated octahedron.

The map is coloured by the area fraction of each family (blue {100}, red {111}, green {110}); the black lines are the exact boundaries above. It treats the two halves of {111} as equal. The tetrahedral shapes that -43m allows when they differ are on the Morphology page.





x = 0.707, y = 1.732: **rhombic dodecahedron**; area {100} 0.000, {110} 1.000, {111} 0.000.

13 What This Shows, and What It Does Not

- **Steps 1 to 7 are exact crystallography of the file.** The cell, the cubic lattice, the centring, the point group, the space group and the effective spacings follow from the numbers in the CIF, and for the four bundled files they agree with spglib.
- **Step 8 is a rule, not a physical model of growth.** BFDH ignores the chemistry of the faces, the solvent, the temperature and any modulator. Step 8 also shows that the BFDH shape depends only on the centring and the point group, so MOF-5, HKUST-1 and UiO-66, all face-centred cubic, receive the same truncated octahedron. Laboratories grow these three MOFs in different shapes, so BFDH cannot by itself tell them apart. How the project's other models (graph attachment numbers, force-field and CHGNet surface energies) move each MOF to a different point of the map in Step 12, and how those points compare with laboratory crystals, is on the Morphology, Surface Energy and Experiment pages.
- **Only equilibrium-like shapes are produced.** Every shape here is a convex polytope with flat faces. Rods, plates and spheres grown under kinetic control or with additives that block particular faces are not produced by this construction.
- **Cubic crystals only.** A file whose lattice is not cubic stops at Step 3 with a message. The Python package handles any lattice, and the Cell Deform page uses it on stretched cells.

> **Where every number comes from**

> Browser: `website/mof_cif_shape.js` (all steps) and `website/cif_shape_page.js` (this page), with `website/mof_morphology.js` for the shape descriptors and names. Python reference: `code/morphology/run_cif_to_shape.py`, which writes `code/morphology/results/cif_to_shape.json`; `website/_build/build_cif_shape_data.py` copies the Python numbers into `website/cif_shape_data.js`. Check: `website/tests/test_cif_to_shape.js`.
